

MMI 891 · CAPSTONE PROJECT

MSc in Mathematical Innovation · Open University of Kenya

Module 5: Project Implementation and Findings

Programme Coordinator | Instructional Hours: 6

LEARNING OUTCOMES

1. Document the project implementation process accurately and systematically, capturing both planned and emergent aspects.
2. Present findings in a clearly structured, professionally formatted manner aligned to each stated objective.
3. Apply appropriate visualisation and tabulation techniques to communicate quantitative and qualitative results.
4. Analyse and interpret results objectively, distinguishing between findings and their interpretation.

1 Module Overview

Chapter Four (Project Implementation and Findings) is the empirical core of your Capstone Project. It demonstrates that you have executed your approved plan and documents *what happened* and *what you found*. This chapter requires disciplined separation between description (what was done), presentation (what the data shows), and interpretation (what it means) — the last of which is properly deferred to Chapter Five (Discussion).

THE FINDINGS/DISCUSSION BOUNDARY

A recurring error in capstone writing is mixing findings and discussion in the same chapter. Chapter Four presents what the data *shows*. Chapter Five explains what the data *means* in relation to existing theory, the research questions, and the project context. In Chapter Four, resist the urge to conclude, theorise, or recommend — focus on accurate, comprehensive, evidenced presentation. Keep the intellectual weight for Chapter Five.

2 Structure of Chapter Four

4.1	Introduction	0.5 page	Chapter overview; link to objectives
4.2	Description of Project Implementation	2–3 pages	What was done; deviations from plan
4.3	Presentation of Findings	5–10 pages	Results per objective; tables, figures
4.4	Analysis and Interpretation of Results	2–3 pages	Patterns; inter-variable relationships

3 Section 4.2: Description of Project Implementation

3.1 Purpose

This section documents what was actually done during implementation — not what was planned (which is in Chapter Three) but what was *executed*. It must account for:

- Each phase and activity completed, with timeline information.
- Any deviations from the approved implementation matrix and their rationale.
- Challenges encountered and how they were resolved.
- Stakeholder engagement activities conducted.
- Any ethical issues encountered during implementation.

3.2 Documenting Deviations

Deviations from the plan are not failures — they are evidence of adaptive practice. Document them honestly:

DOCUMENTING IMPLEMENTATION DEVIATIONS

What changed: State what activity, timeline, or method changed from the approved plan.

Why it changed: State the reason — data access delays, participant unavailability, unexpected data quality issues, system constraints.

Impact on objectives: Assess whether the deviation affects the project's ability to meet its stated objectives.

Mitigation: Describe what was done to preserve the project's integrity despite the deviation.

EXAMPLE: Implementation Description Excerpt

Phase 1 (data preparation) was completed in Month 1, one week ahead of schedule. The OUK SIS data extract was received on 5 March 2024 and contained 8,743 student records. Following application of exclusion criteria (records with >30% missing predictor variables), the analytical dataset comprised 7,218 records — slightly above the projected 7,100. EDA was completed by 15 March 2024.

Deviation 1: The LMS engagement log data (planned as a feature source) was unavailable due to a system migration at OUK that commenced unexpectedly in February 2024. This data was therefore excluded from the feature set. Impact: two planned engagement-based features (login frequency, resource access rate) could not be computed. Mitigation: additional academic performance features (assignment submission rate, class test scores) were derived from the SIS as replacement predictors, maintaining the planned feature count.

Phase 2 (model development) proceeded as planned in Months 2–3. All 12 planned interviews for Phase 3 were completed by the end of Month 5; two participants were replaced due to change of employment, with permission from the ethics committee to recruit substitutes from the same role categories.

4 Section 4.3: Presentation of Findings

4.1 Organisation by Objective

Findings must be presented by objective — each research objective generates findings that are directly reported here. Use the following structure for each:

FINDINGS PRESENTATION STRUCTURE PER OBJECTIVE

Objective [n] heading: Restate the objective.

Finding statement: State the key result concisely.

Evidence: Present the data — tables, figures, model outputs, statistical results, or thematic results — supporting the finding.

Description: Describe what the data shows (do not interpret yet).

4.2 Quantitative Findings: Tables and Figures

TABLE AND FIGURE STANDARDS

Tables:

- Every table has a number (Table 4.1, Table 4.2, ...) and a descriptive title above it.
- All columns and rows are clearly labelled with units.

- Source noted below (“Author’s own analysis, 2024” or “Adapted from ...”).
- Referenced in the text before the table appears.

Figures:

- Every figure has a number (Figure 4.1, Figure 4.2, ...) and a descriptive caption below it.
- Axes are labelled with units. Legends are included where multiple series appear.
- Resolution: minimum 300 dpi for submission.
- Referenced in the text before the figure appears.

EXAMPLE: Quantitative Findings Presentation

Objective 2: To develop a machine learning classification model for predicting first-semester dropout risk.

Finding: The Random Forest classifier (RF) achieved the highest predictive performance with an AUC-ROC of 0.83 on the test set, outperforming the Logistic Regression baseline (AUC = 0.71).

Table 4.3: Model Performance Comparison on the Test Set ($n = 1,082$)

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.78	0.61	0.57	0.59	0.71
Random Forest	0.84	0.76	0.72	0.74	0.83
Random Forest (tuned)	0.86	0.79	0.76	0.77	0.85

Source: Author’s own analysis, 2024. Bold indicates best-performing model.

As shown in Table 4.3, the tuned Random Forest model outperformed both baseline classifiers across all evaluation metrics. The recall of 0.76 indicates that the model correctly identifies 76% of actual dropout cases — a critical performance indicator for an early-warning application where missed cases (false negatives) have the greatest operational cost.

4.3 Qualitative Findings: Thematic Presentation

EXAMPLE: Qualitative Findings Presentation

Objective 4: To propose an operationally feasible early-warning system protocol for OUK’s Student Affairs Division.

Finding: Three themes emerged from the 12 stakeholder interviews regarding early-warning system design requirements: (1) *Alert timeliness* — staff require predictions in the first two weeks of semester; (2) *Actionable outputs* — raw probability scores are insufficient; staff need flagged intervention categories; (3) *Workload integration*

— the system must integrate with existing Student Affairs workflows, not require a separate login.

Theme 1: Alert Timeliness

Ten of 12 participants identified the timing of at-risk identification as the most critical design feature. As Participant 3 (Programme Coordinator) noted: “By the time we get end-of-semester data, it’s too late. We need to know in Week 2, not Week 14.” This was consistent with findings from the quantitative model, which showed that first-two-week engagement variables were among the top-5 predictors by feature importance (see Figure 4.4).

4.4 Feature Importance and Model Interpretability

For machine learning projects, present feature importance or SHAP value analysis to explain model outputs:

EXAMPLE: Feature Importance Reporting

Figure 4.4: Top 10 Predictors by Random Forest Feature Importance

[Insert bar chart with features on y-axis and importance score on x-axis]

Figure 4.4 shows that the three most important predictors are: (1) first-semester GPA (importance score: 0.21); (2) fee payment status at Week 2 (0.18); and (3) assignment submission rate in Weeks 1–2 (0.15). These findings are consistent with Bean and Metzner’s (1985) model predicting that financial access and early academic engagement are primary retention determinants.

5 Section 4.4: Analysis and Interpretation of Results

In this section, move from describing individual findings per objective to analysing patterns *across* objectives. This section addresses:

- Convergence: where quantitative and qualitative findings reinforce each other.
- Divergence: where findings from different strands appear contradictory, and what might explain the difference.
- Unexpected findings: results that were not anticipated by the conceptual framework.
- Cross-objective synthesis: patterns that emerge when findings across multiple objectives are considered together.

EXAMPLE: Cross-Strand Analysis

A notable pattern of convergence emerged between the quantitative model findings and the qualitative stakeholder data. The quantitative analysis identified fee payment status and early assignment submission as the two strongest predictors of dropout (Section 4.3.2). The qualitative data independently surfaced financial ac-

cess as the most consistently cited barrier to student persistence (Theme 1, Section 4.3.4). This convergence across independent data strands strengthens confidence that financial vulnerability is a primary, rather than secondary, dropout driver at OUK.

An unexpected finding was the limited predictive contribution of prior academic qualification (school leaving grade) relative to in-semester academic behaviour variables. This diverges from the majority of existing literature, which reports prior attainment as a strong predictor (see Section 2.2). A possible explanation is that OUK's flexible entry requirements reduce the variance in prior qualification, weakening its discriminative power in this specific context.

6 Common Chapter Four Errors

TEN CRITICAL ERRORS IN FINDINGS CHAPTERS

1. Presenting discussion/conclusions in the findings chapter.
2. Tables and figures not referenced in the text before they appear.
3. Missing table titles or figure captions.
4. Reporting percentages without stating the base (n).
5. Stating that a result is “significant” without specifying the test and p -value.
6. Reporting qualitative findings without supporting participant quotes.
7. Using participant names instead of pseudonyms or participant codes.
8. Presenting only favourable results; omitting unexpected, null, or contradictory findings.
9. Findings not organised by objective — the reader cannot tell which finding addresses which objective.
10. Model performance reported only as accuracy without recall, precision, and AUC.

7 Module Assignment

MODULE 5 ASSIGNMENT

Deliverable: Submit the Project Implementation and Findings chapter in accordance with OUK approved format.

Checklist:

1. Section 4.1: Introduction linking chapter to objectives.
2. Section 4.2: Implementation description — all phases documented; deviations reported with justification; timeline adherence assessed.

3. Section 4.3: Findings per objective — each objective has a dedicated findings section; minimum 3 tables and 2 figures; qualitative findings include participant quotes (pseudonymised); all tables/figures numbered, titled, sourced, and referenced in text.
4. Section 4.4: Cross-strand analysis — at least one convergence, one divergence, and one unexpected finding discussed.

Format: 15–20 pages. APA 7th throughout. All figures at minimum 300 dpi.

8 Revision Questions

1. A student reports: “The model accuracy was 85%, which is significant.” What is wrong with this statement? Rewrite it as a properly reported quantitative finding.
2. Explain the boundary between findings (Chapter Four) and discussion (Chapter Five). Give an example of a sentence that correctly belongs in each chapter.
3. A machine learning model for drought prediction achieves 92% accuracy but only 31% recall on the drought class. Explain what this means in practical terms for a Kenyan county water manager relying on the system.
4. Your planned sample of 300 survey respondents yielded only 187 completed responses. How should you document this deviation, and how does it affect your findings chapter?
5. How should unexpected findings be treated in the findings chapter? Is it acceptable to omit them? Justify.
6. For a systems dynamics model of urban water supply: describe what Figure 4.1 (stock-and-flow diagram) and Table 4.1 (scenario comparison results) should contain, and write the caption and reference text for each.
7. What is SHAP analysis and why is it important for interpreting machine learning models in applied capstone projects?

MODULE 5 SUMMARY

- Chapter Four presents implementation and findings — what was done and what the data shows — strictly separated from interpretation (Chapter Five).
- The implementation section must document both successes and deviations from the plan, with justification and impact assessment.
- Findings must be organised by objective, presented with appropriate tables and figures, and supported by evidence (statistics or participant quotes).
- Cross-strand analysis (Section 4.4) identifies convergence, divergence, and unexpected findings across the project’s evidence base.
- All tables and figures must be numbered, titled/captioned, sourced, and referenced in the text.
- *Your findings chapter is your evidence — treat it with the rigour and precision of a legal brief: claim nothing you cannot immediately support with data.*

9 Further Reading

- **Creswell, J.W.** (2023). *Research Design* (6th ed.). SAGE. — Chapter 12: Writing a Mixed Methods Study.
- **Lundberg, S. & Lee, S.I.** (2017). A unified approach to interpreting model predictions. *NeurIPS 2017*. (SHAP values for model interpretability.)
- **American Psychological Association.** (2020). *Publication Manual* (7th ed.). — Chapter 7: Tables and Figures.